Graph Modelling and Spatial-Explicit GeoAI for Urban Analytics

刘鹏远, Pengyuan Liu
Urban Analytics Subject Group
Urban Studies & Social Policy Division
Urban Big Data Centre
University of Glasgow

About me

刘鹏远

Dr. Pengyuan Liu

Education Background：

PhD in Geography

University of Leicester, UK

MSc in Cloud Computing

University of Leicester, UK

BSc in Network Engineering

Nanjing University of Post and Telecommunication, China

Employment：

Lecture in Digital Planning

University of Glasgow

Module Coordinator – Future Cities Lab Global

Singapore-ETH Centre

Lecturer – Department of Geographical Sciences

Nanjing University of Information Science and Technology

Postdoc Research Fellow – Urban Analytics Lab

National University of Singapore

Postdoc Researcher – Digital Geography Lab

University of Helsinki

Two Sessions

- **Session 1 — Graph Modelling for Urban Analytics**
  Constructing urban graphs (nodes/edges), encoding spatial relationships, and preparing graph-structured data for learning.

- **Session 2 — Spatial-Explicit GeoAI with Graph Neural Networks**
  Applying GNNs to spatial prediction tasks and discussing uncertainty, generalisation, and interpretability in urban settings.

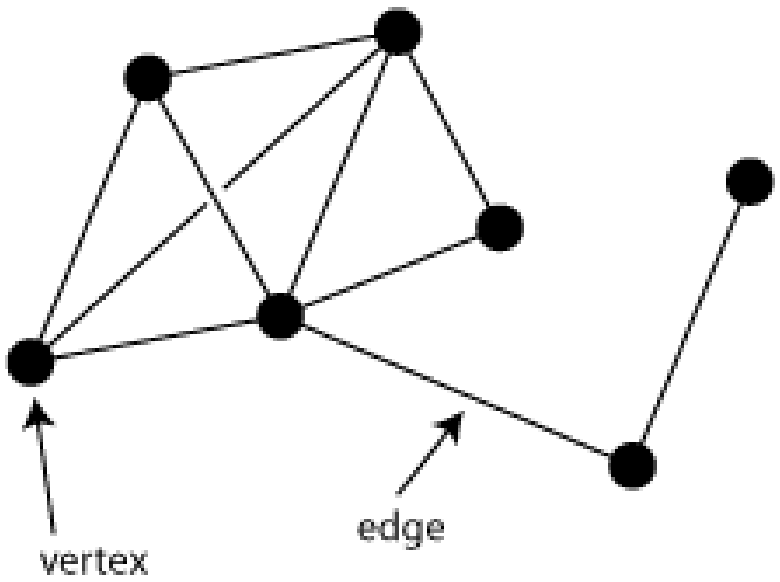
Hey, what is a *Graph*?

Introduction to Graph

We are not differentiating the concept between network and graph in this course!

A graph is a way to represent objects and the relationships between them.

- Objects → Nodes (Vertices)
- Relationships → Edges (Links)



graph

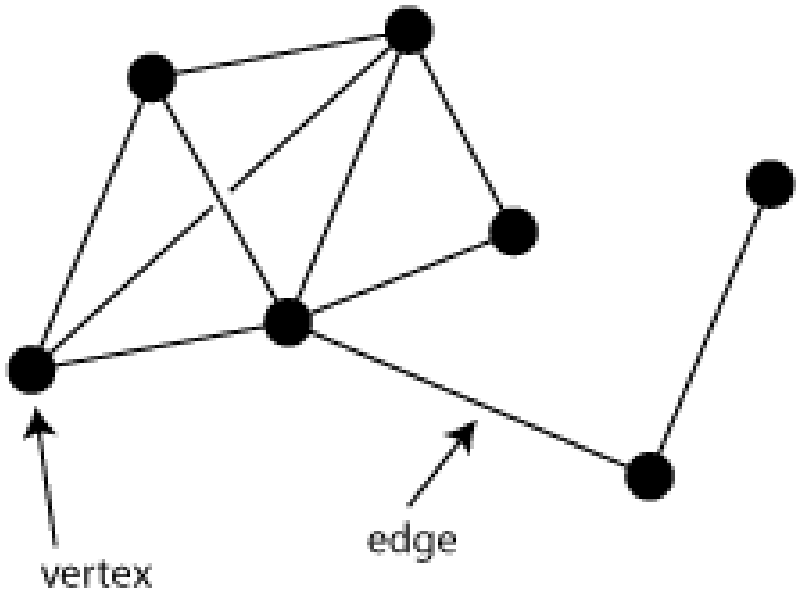
Introduction to Graph

Formally, a graph is defined as:

$$G = (V, E)$$

Where:

- V = set of nodes
- $E \subseteq V \times V$ = set of edges



graph

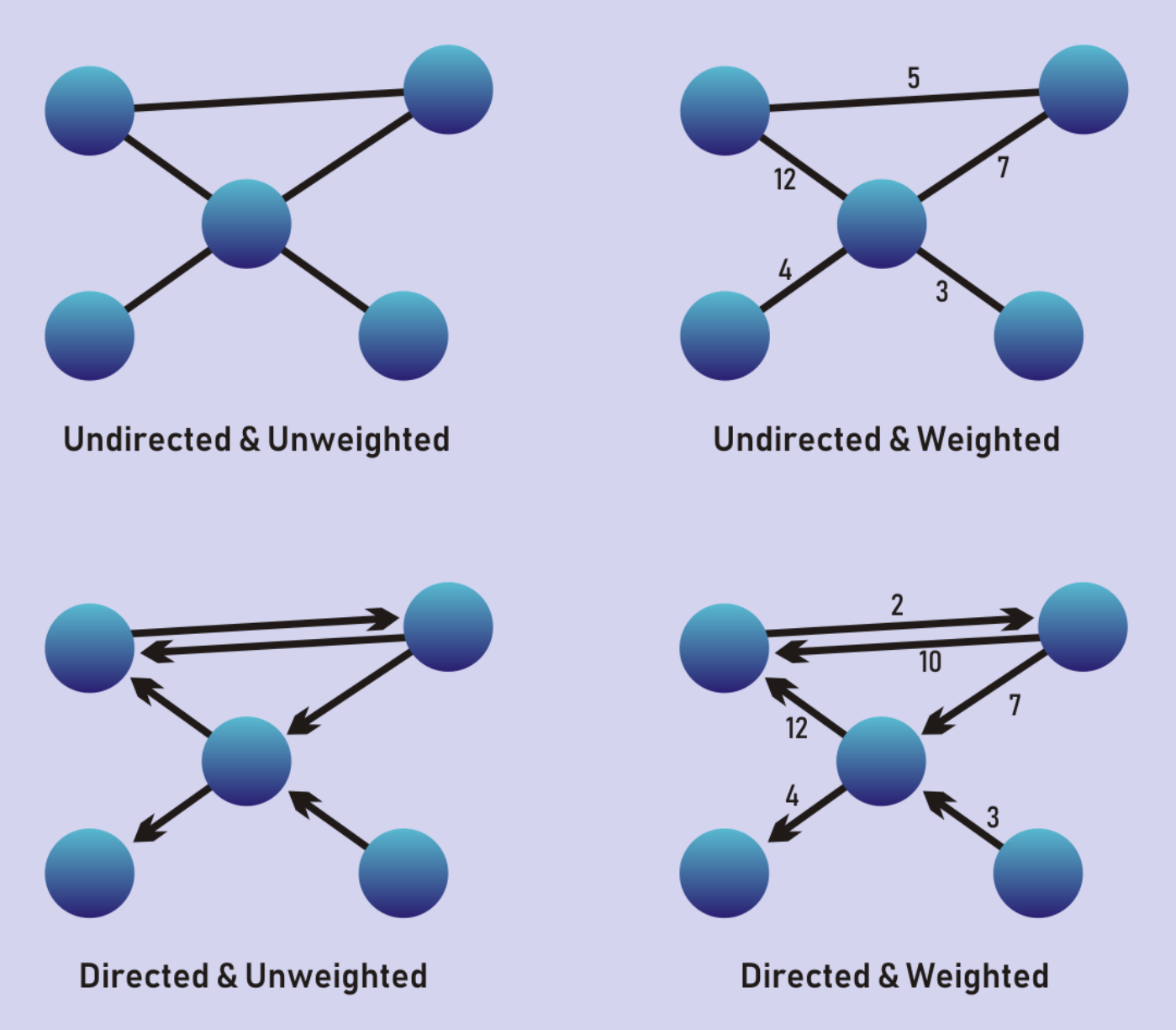
Introduction to Graph

Formally, a graph is defined as:

$$G = (V, E)$$

Where:

- V = set of nodes
- $E \subseteq V \times V$ = set of edges



Source: https://medium.com/tebs-lab/types-of-graphs-7f3891303ea8

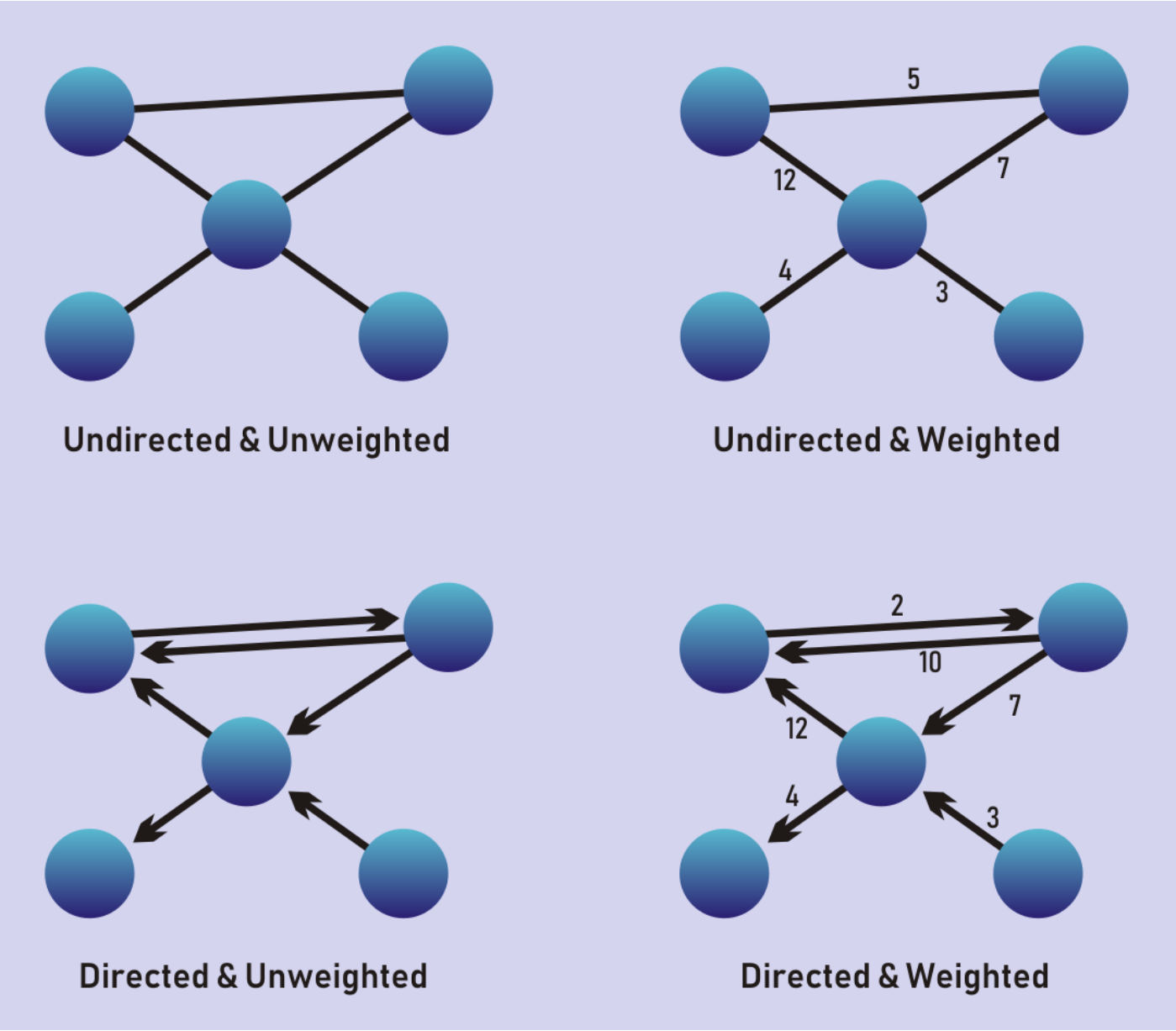
Introduction to Graph

For a graph $G = (V, E)$, the adjacency matrix is:

$$A_{ij} = \begin{cases} w_{ij} & \textit{if node } i \textit{ is connected to node } j \\ 0 & \textit{otherwise} \end{cases}$$

For unweighted graph, $w_{ij} = 1 \; or \; 0$;

For weighted graph,
w_{ij} depends on the weight values



Source: https://medium.com/tebs-lab/types-of-graphs-7f3891303ea8

Fundamental Graph Properties

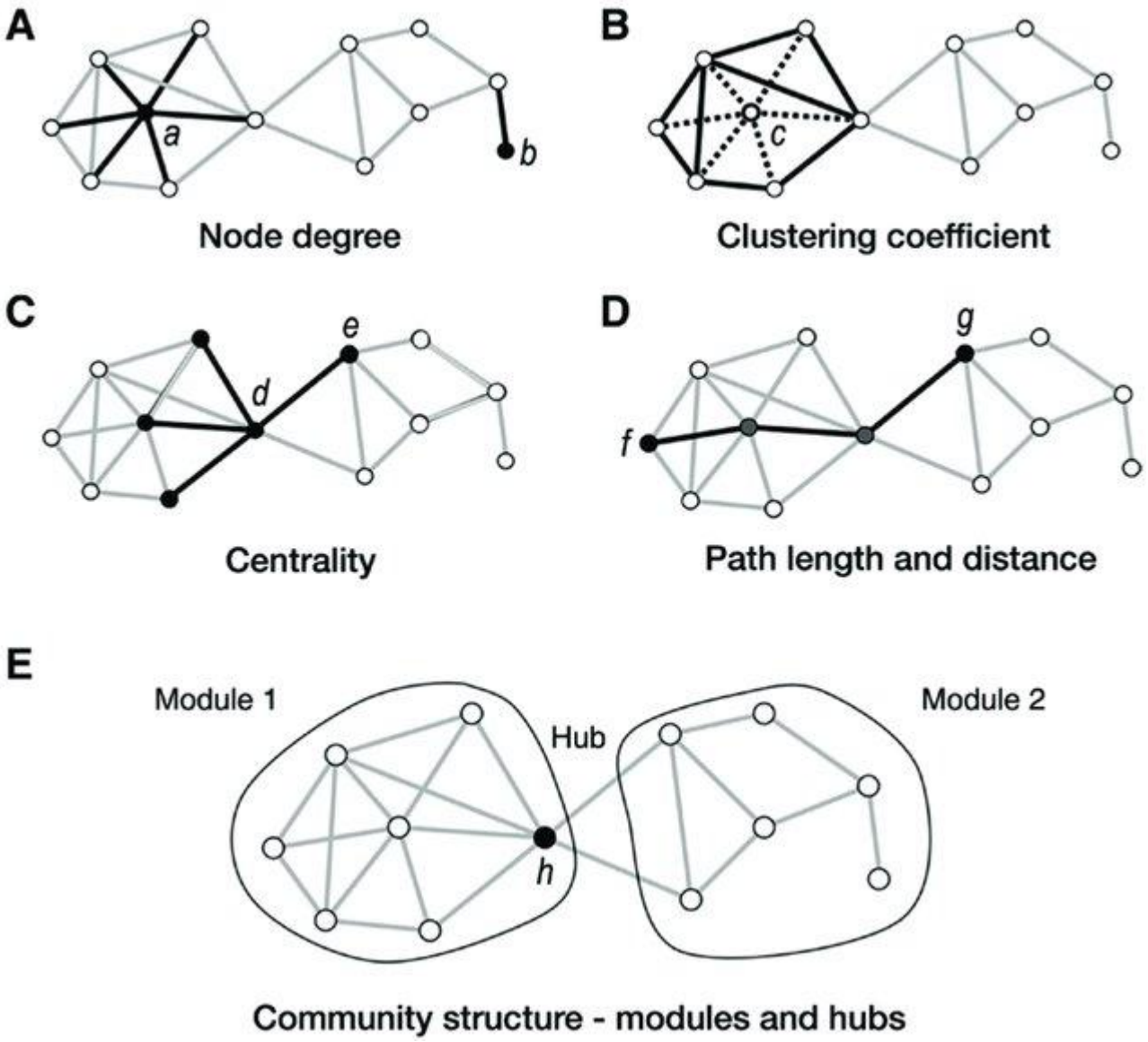
(A) Degree: the number of edges attached to a given node. The node *a* has a degree of 6, and the peripheral node *b* has the degree of 1.

(B) Clustering coefficient: the extent to which nodes tend to cluster together, measuring the segregation of a network. In this example, the central node *c* has 6 neighbours and 15 possible connections among the 6 neighbours. These neighbours maintain 8 out of 15 possible edges. Thus, the clustering coefficient is 0.53 (8 of 15).

(C) Centrality: the indicators of centrality identify the most influential nodes within a network. In a social network, it is used to identify the most influential person. In this example, node *d* contributes more to the centrality because all nodes on the right side pass through the node *d* to reach the other nodes in the left side.

(D) Path length: The average of the shortest distances for all node pairs in a network. The shortest path length between the nodes f and g is three steps that pass through two intermediate nodes.

(E) Modularity: one measure of the structure of networks that is designed to reflect the strength of a division of a network into modules (also called groups, clusters, or communities).



Lee, U. and Mashour, G.A., 2018. The role of network science in the study of anesthetic state transitions. *Anesthesiology*, *129*(5), p.1029.

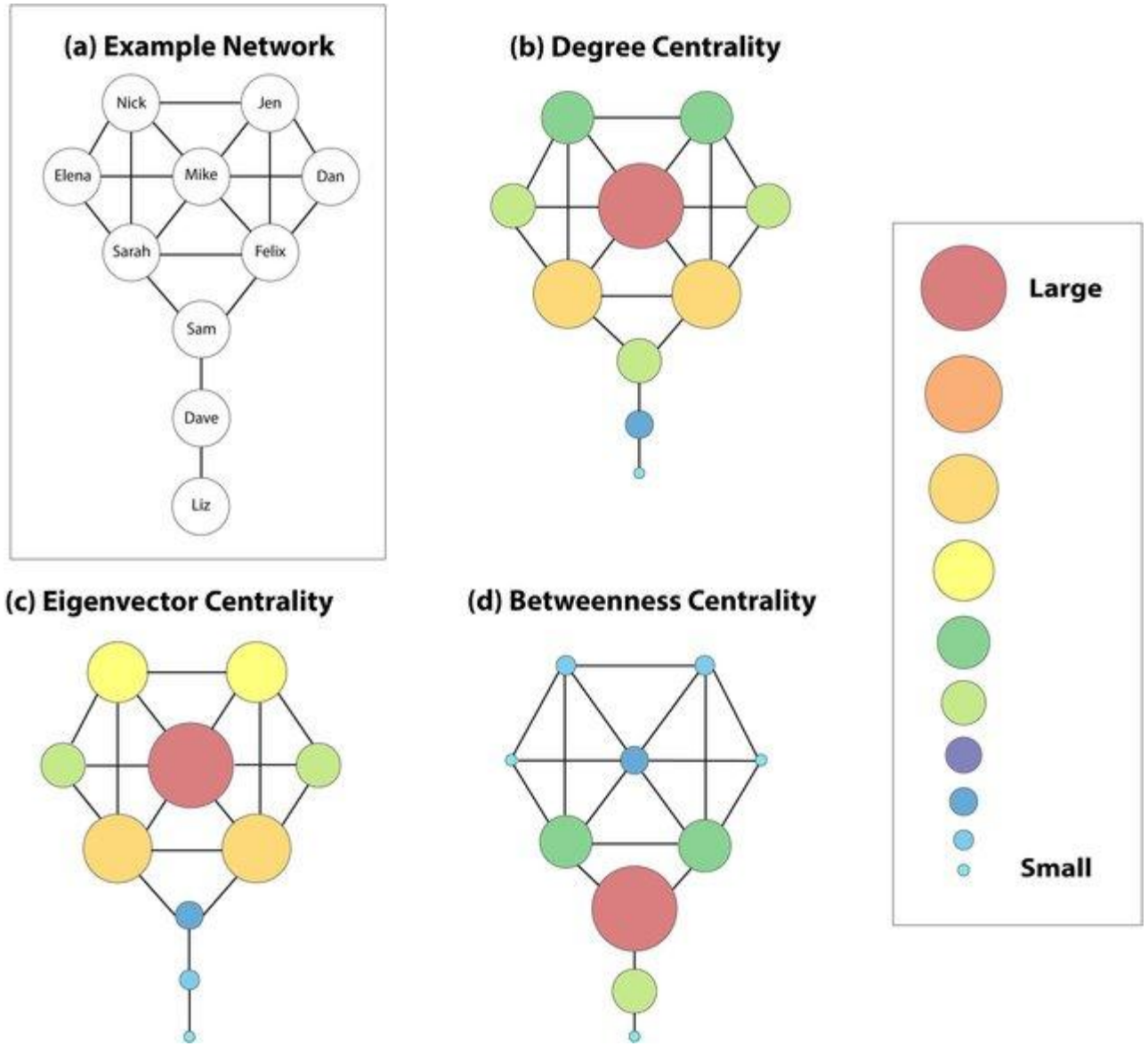
Centrality

(A) An example friendship network, with each node labelled with the name of an individual. (B-D) Variations of the same network, with the nodes resized to reflect the value of a particular centrality measure.

(B) Degree centrality (i.e. degree) is the number of other nodes to which a node is connected directly (i.e. adjacent). Mike has a degree of 7, the largest value in the network.

(C) Eigenvector centrality captures how well-connected a node is to well-connected others. Although Elena, Dan, and Sam all have the same degree (it is 3), Sam has a much smaller eigenvector centrality, as his friendships are with relatively poorly connected individuals.

(D) Betweenness centrality captures the extent to which a node lies on shortest paths between pairs of nodes. Sam has the largest betweenness centrality in this network, because he connects many nodes in the network that otherwise would be on disconnected components of the network.



Baek, E.C., Porter, M.A. and Parkinson, C., 2021. Social network analysis for social neuroscientists. *Social Cognitive and Affective Neuroscience*, *16*(8), pp.883-901.

Basic Graph Types

Path Graph P_n

A linear chain of vertices where each node connects to at most two neighbours; represents the simplest connected topology without cycles.

Cycle Graph C_n

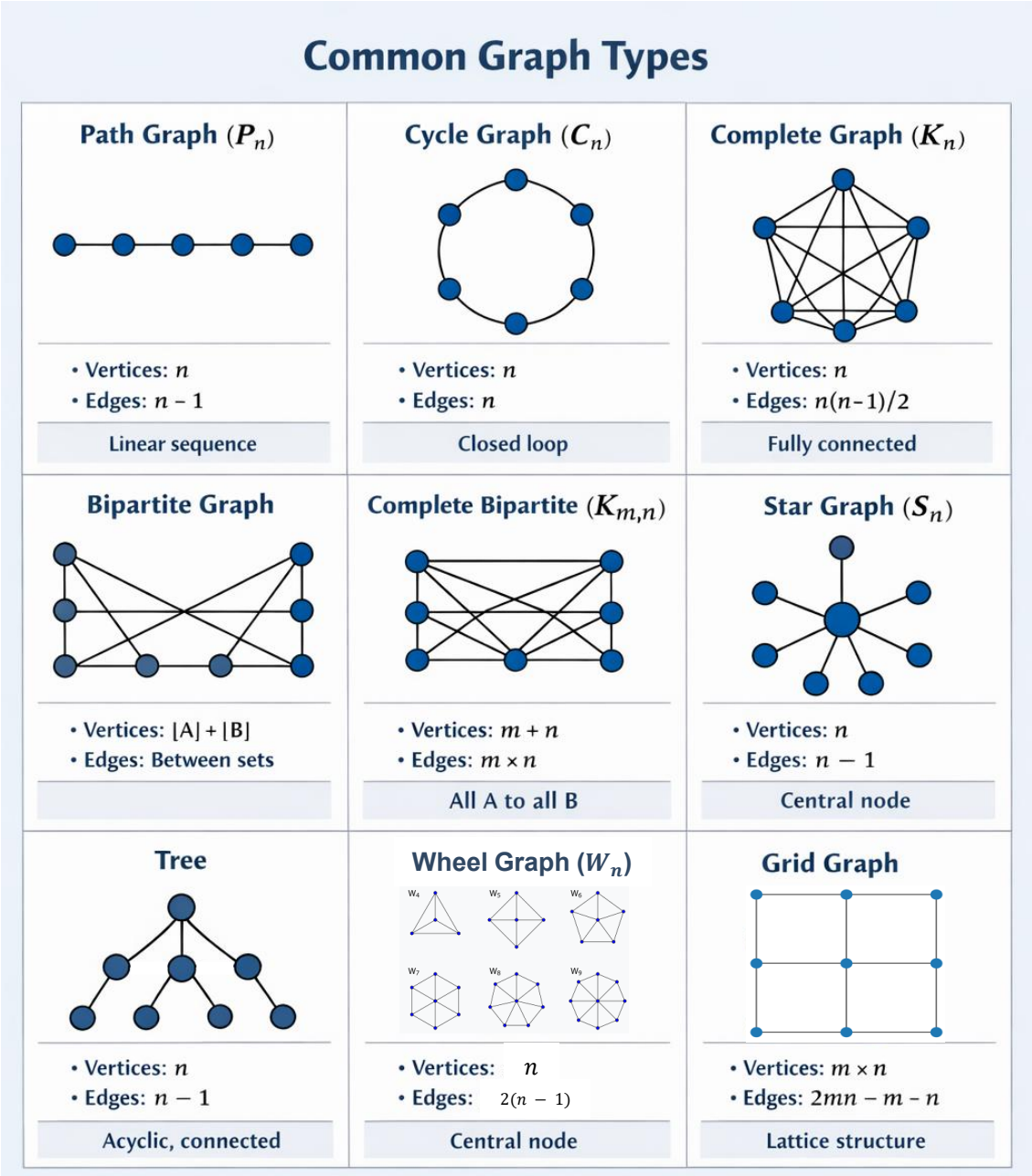
A closed-loop extension of a path graph in which all vertices have degree 2; introduces periodic connectivity.

Complete Graph K_n

A maximally connected graph where every pair of vertices shares an edge; represents the upper bound of network density.

Bipartite Graph

Vertices are partitioned into two disjoint sets with edges only across sets; useful for modelling relational systems (e.g., agent–resource).



Basic Graph Types

Complete Bipartite Graph $K_{m,n}$

A fully connected bipartite structure linking every vertex in set A to every vertex in set B.

Star Graph S_n

A special tree with one central hub connected to all peripheral nodes; exhibits strong degree centralisation.

Tree

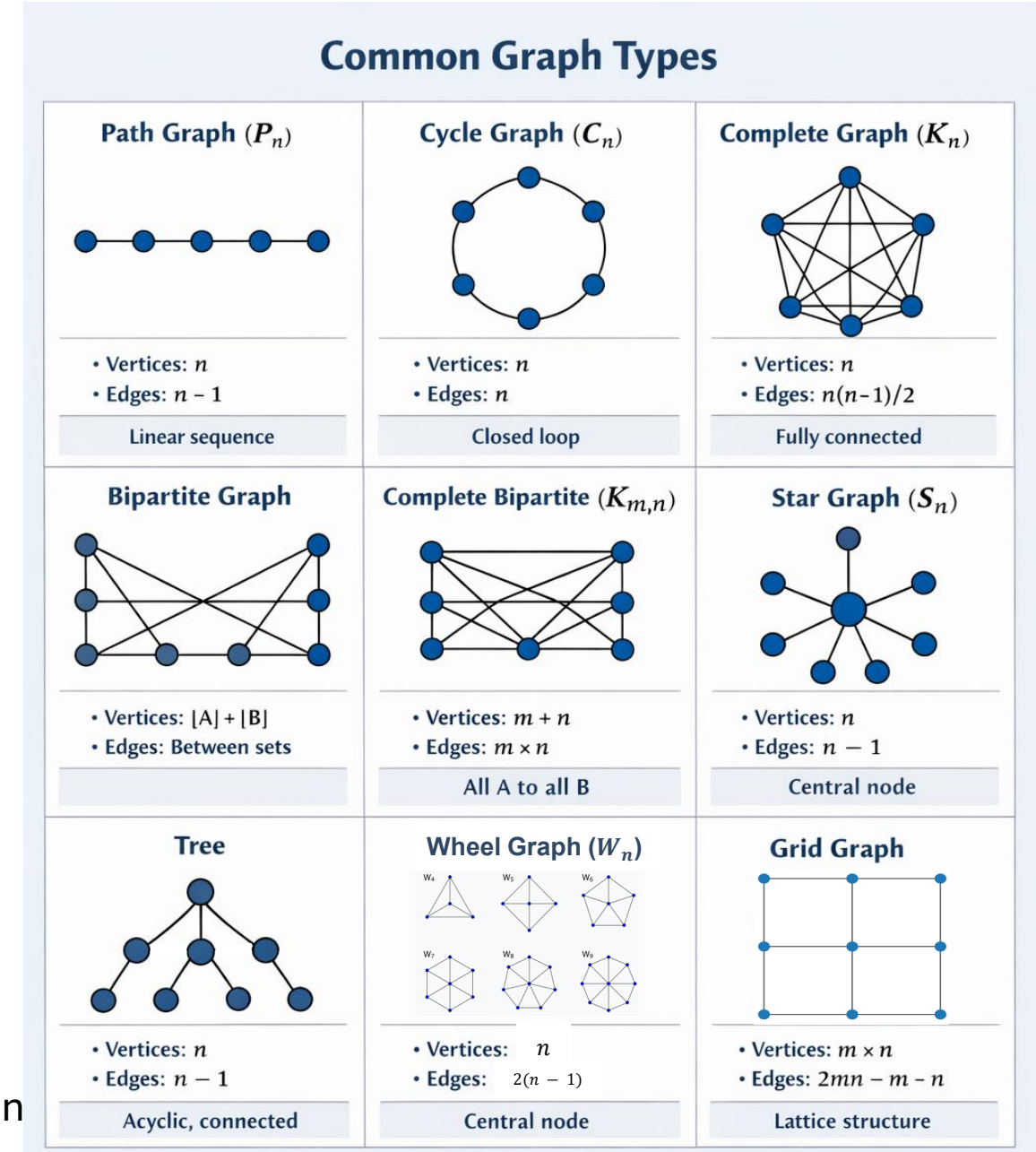
A connected acyclic graph with a unique simple path between any two vertices; minimal edge structure for connectivity.

Wheel Graph W_n

A cycle graph augmented with a central hub node connected to all peripheral vertices.

Grid Graph Q_n

A lattice-like topology formed by regularly arranged vertices; often used to approximate spatial adjacency (e.g., raster surfaces).

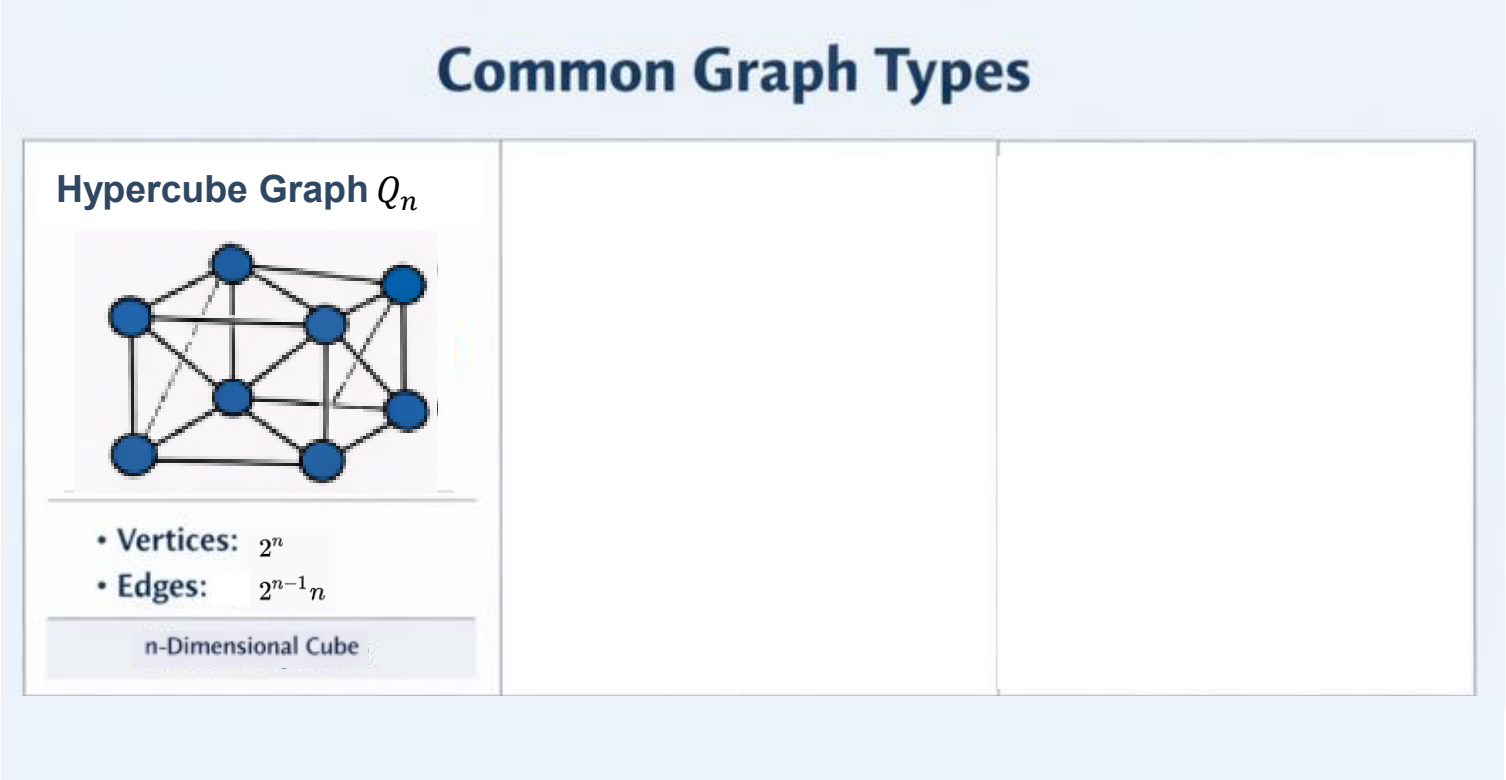


Basic Graph Types

Hypercube Graph Q_n

Vertices represent all binary strings of length n.

Two vertices are connected if they differ in exactly one bit.

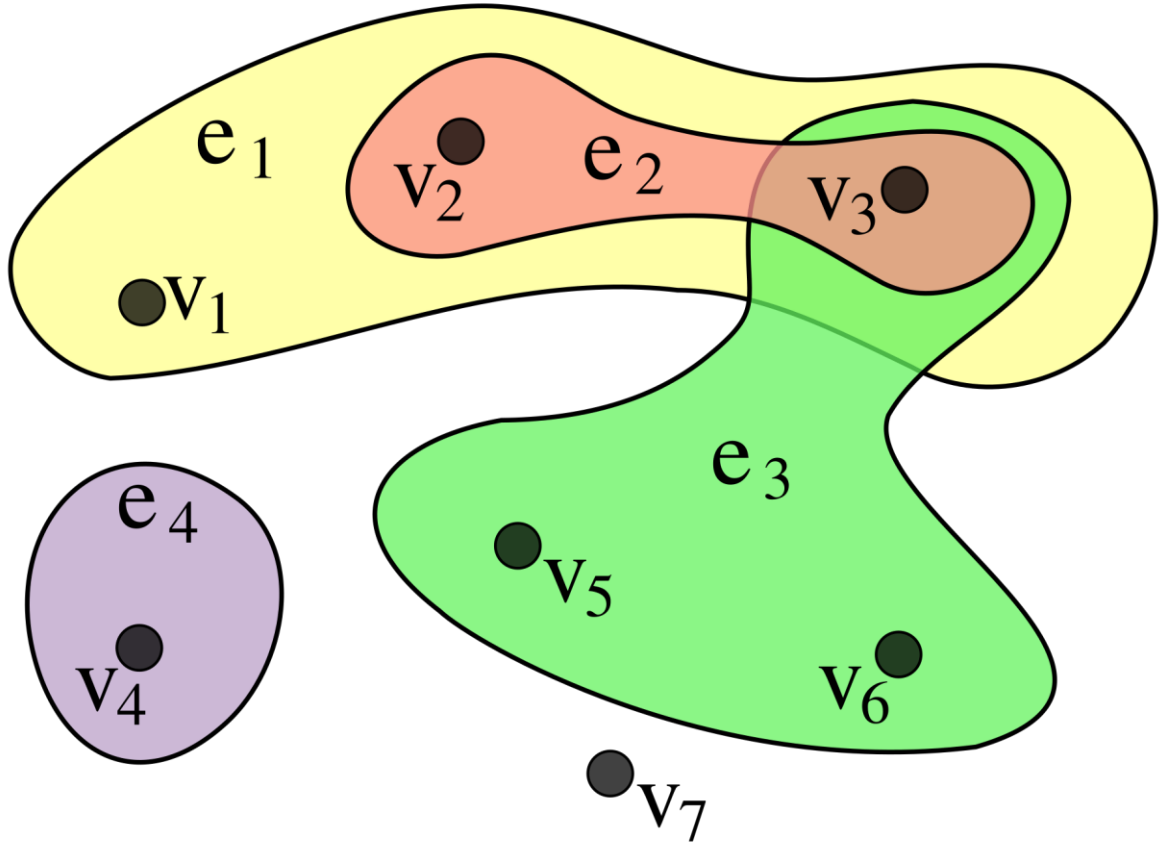


An Advanced Graph Type

Hypergraph

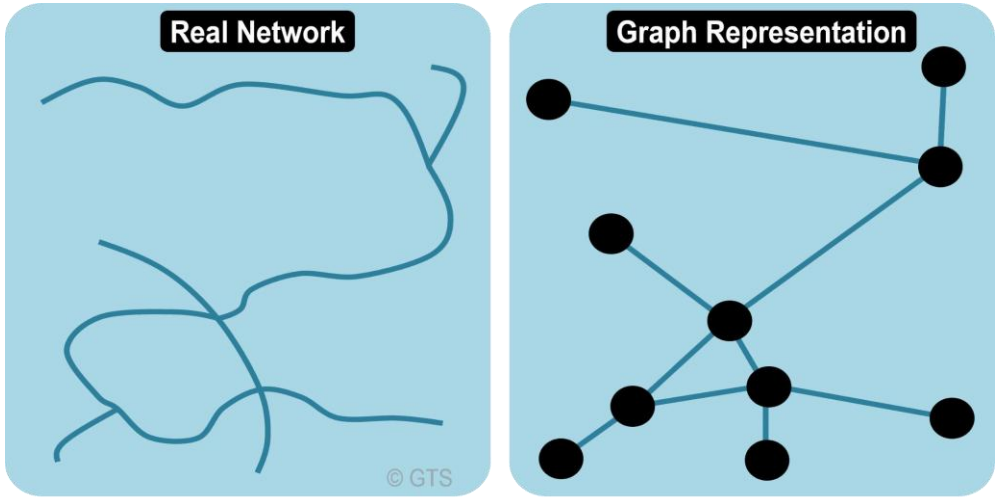
A hypergraph generalises a graph by allowing an edge (hyperedge) to connect more than two vertices.

$X = \{v_1, v_2, v_3, v_4, v_5, v_6, v_7\}$ and $E = \{e_1, e_2, e_3, e_4\} = \{\{v_1, v_2, v_3\}, \{v_2, v_3\}, \{v_3, v_5, v_6\}, \{v_4\}\}$.



Okay, enough for the theory part, why that matters to us??

To be fair, graphs have been really useful



From a geography perspective….

Graphs have long been used in geography and GIScience

- to represent networks
  - transportation networks
  - street networks (geographic)
  - space syntax
  - social networks
- to encode proximity
  - distance weights

Contains National Statistics data Crown copyright and database right 2015; Contains Ordnance Survey data Crown copyright and database right 2015. Data by OpenStreetMap, under ODbL, and by Boeing (2020), under CC0 1.0.

But, why is that?

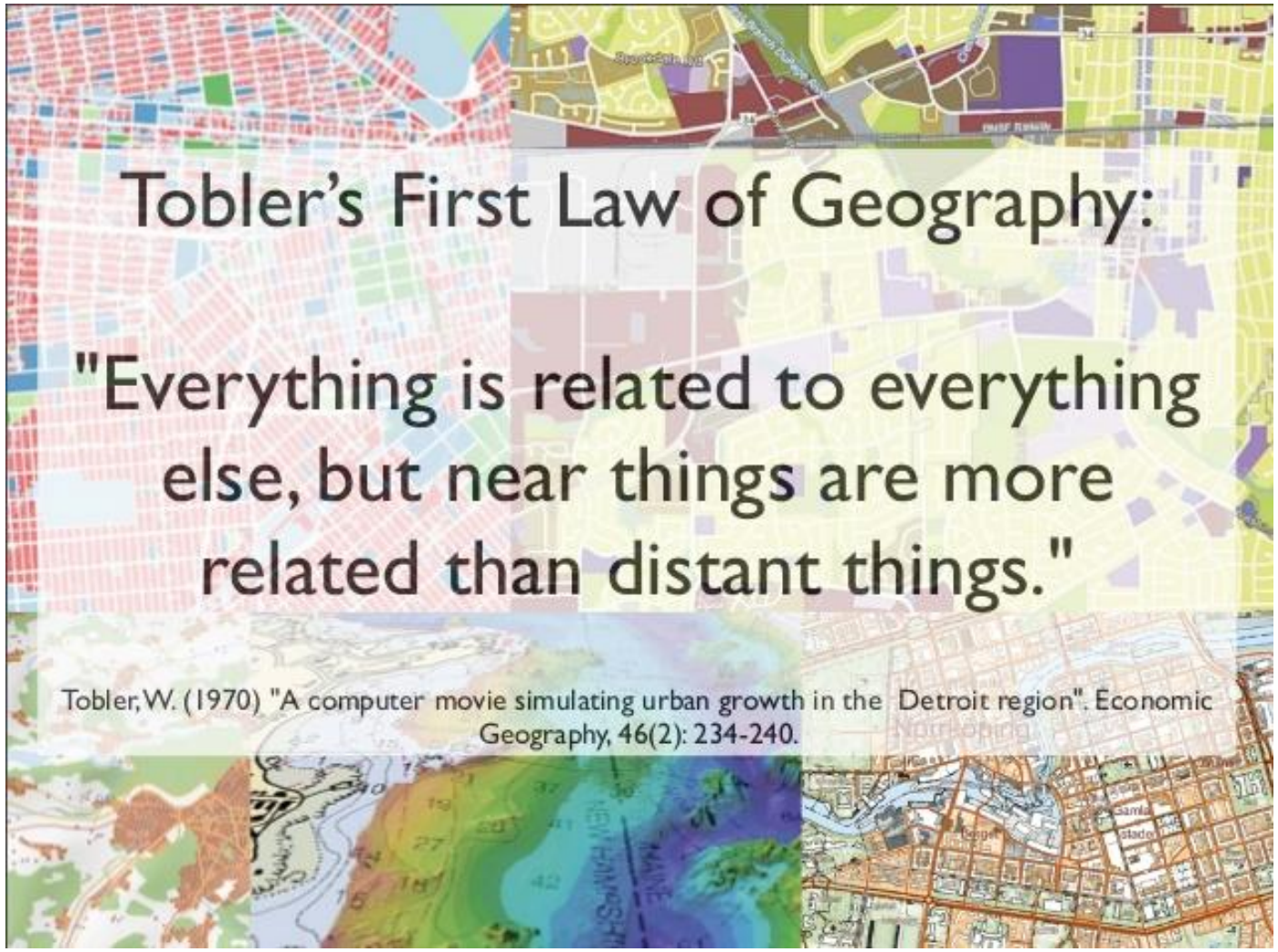
Spatial thinking!

- Location influences almost everything
  - Housing prices depend on where properties are
  - Pollution, noise, heat vary across space
  - Services (schools, parks, transport) are spatially uneven
  - People's behaviours and decisions are shaped by geography

In cities, "where" is just as important as "what."

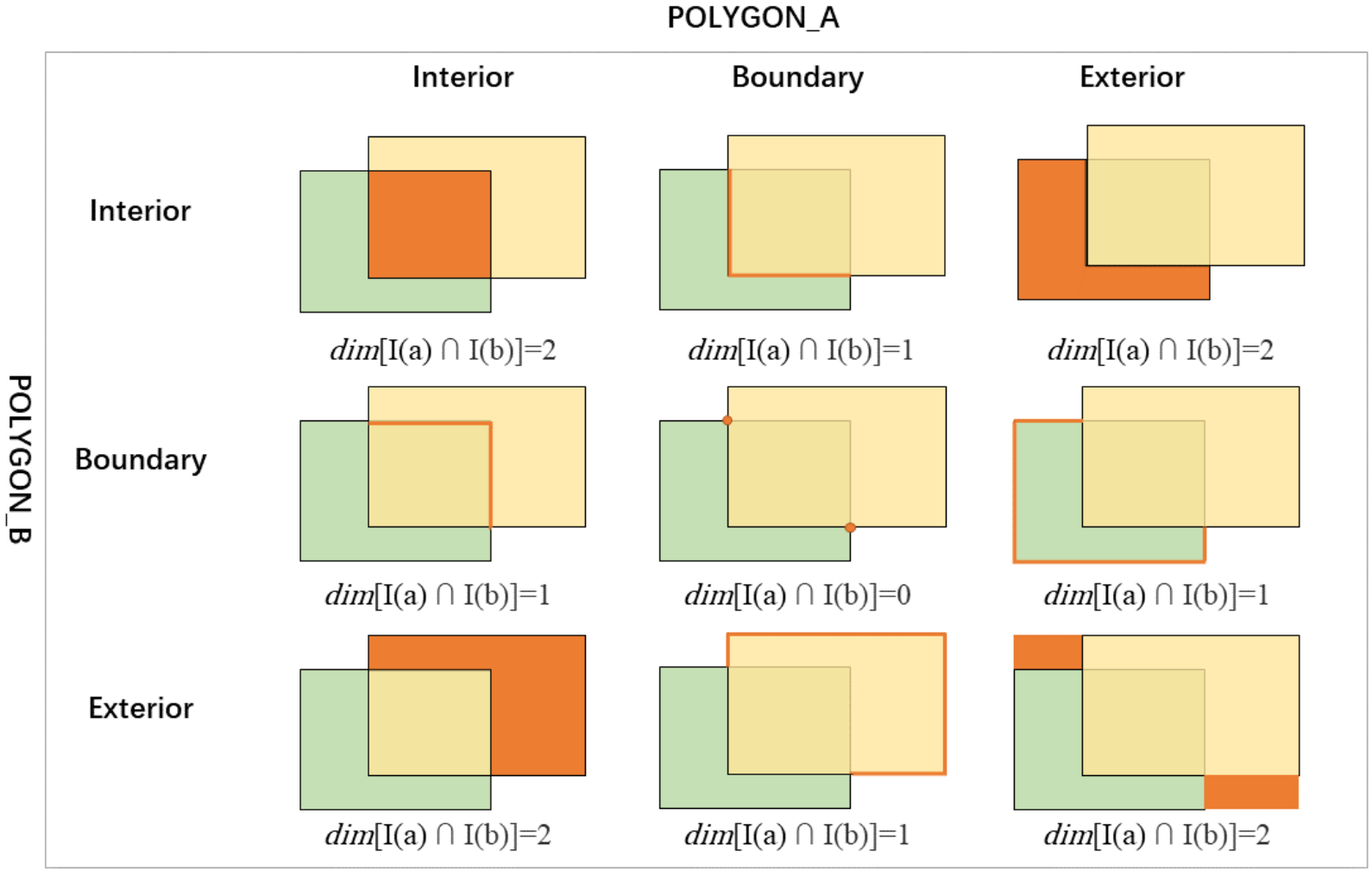
- Traditional statistics ignore space
  - Regression assumes observations are independent
  - But in cities, nearby places influence each other
  - Violates a core assumption → misleading results

Example:
"Two neighbouring houses often have more similar prices than two far-apart houses."



Meanwhile, real-world spatial data are “unstructured”

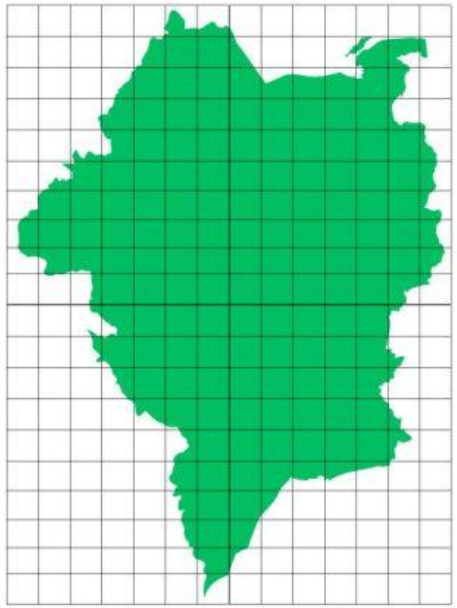
The **Dimensionally Extended 9-Intersection Model (DE-9IM)** is a topological model and a standard used to describe the spatial relations of two regions

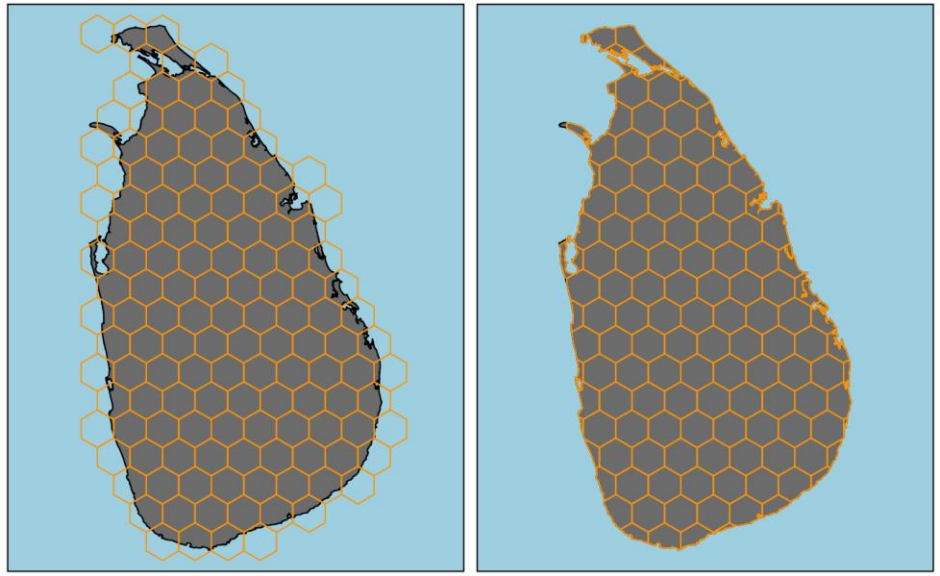


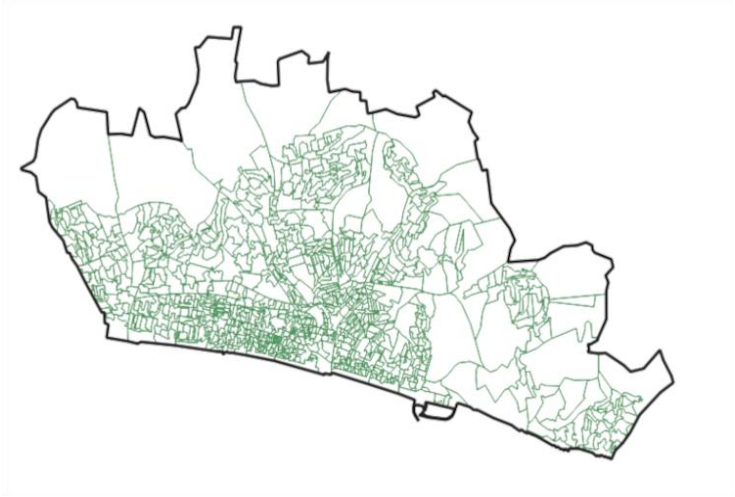
Source: https://yukon.supermap.io/2.0.1/demos/demo_de9im.html

Meanwhile, real-world spatial data are “unstructured”

Spatial Units can be in all forms! Our real-world has never be a “LEGO Land”

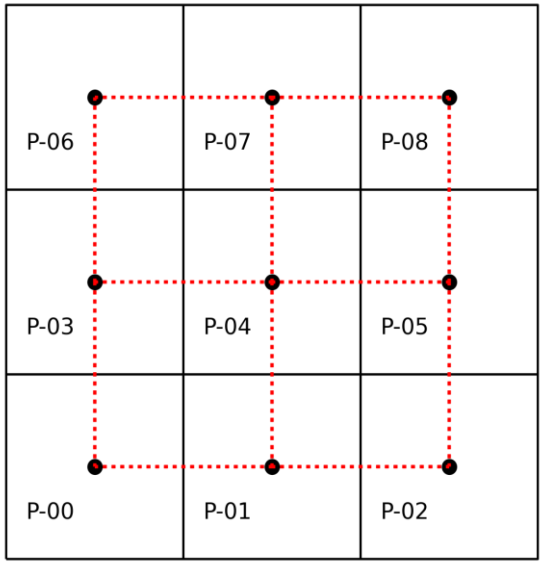




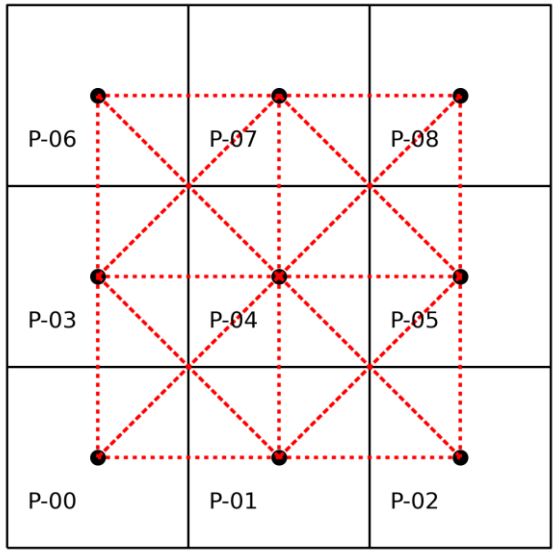


Thus, a high-level understanding of the spatial connections among spatial units is important!

Rook Spatial Weights



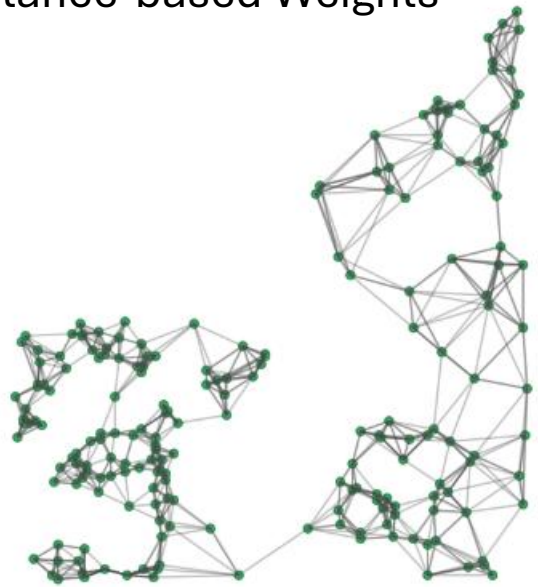
Queen Spatial Weights



Queen

K-NN 4

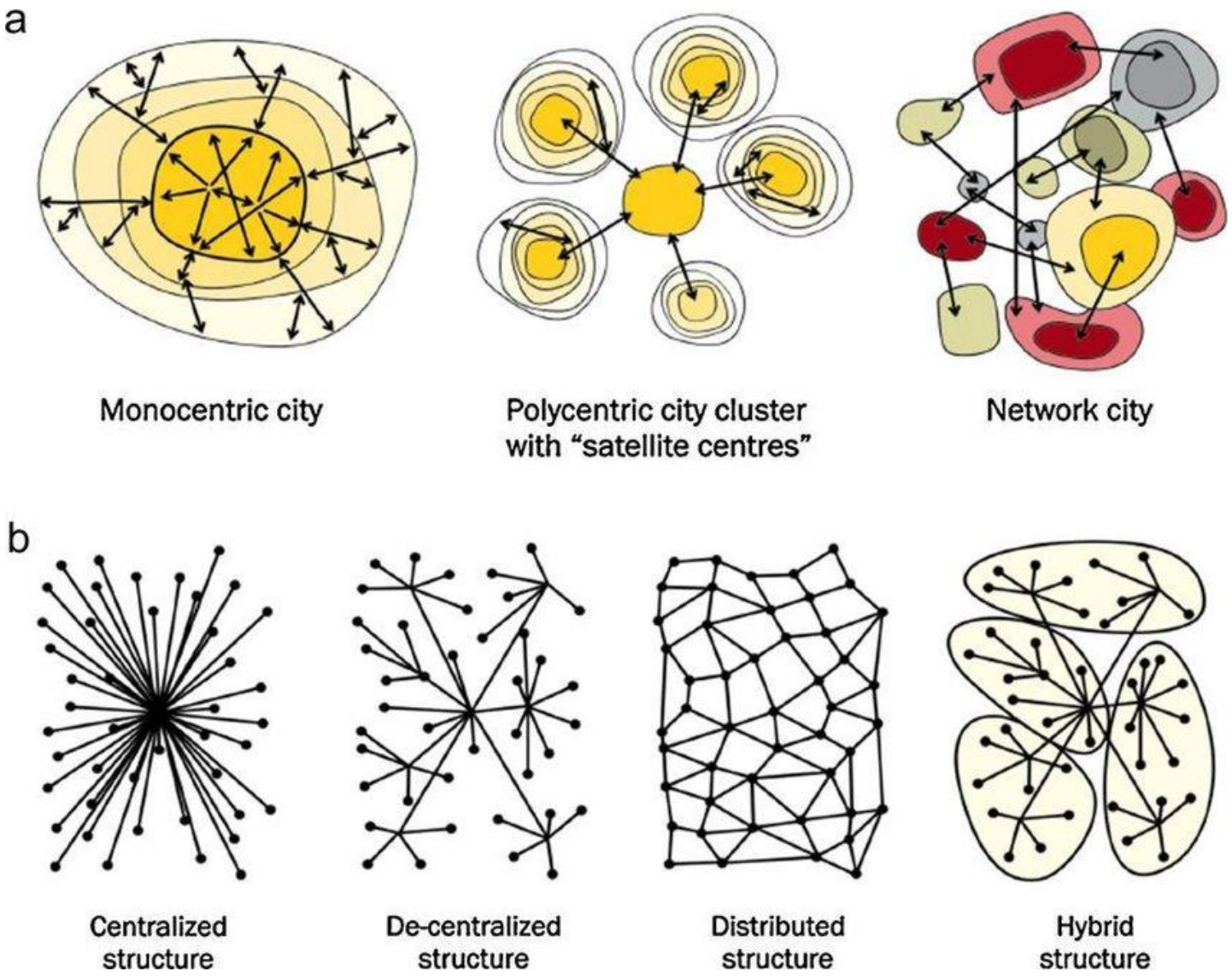
Distance-based Weights



Source:
https://geodacenter.github.io/workbook/4b_dist_weights/lab4b.html
https://geographicdata.science/book/notebooks/04_spatial_weights.html

Essentially, Urban areas are spatial units and following the rules of geography!

- For decades, cities are understood through networks and graphs



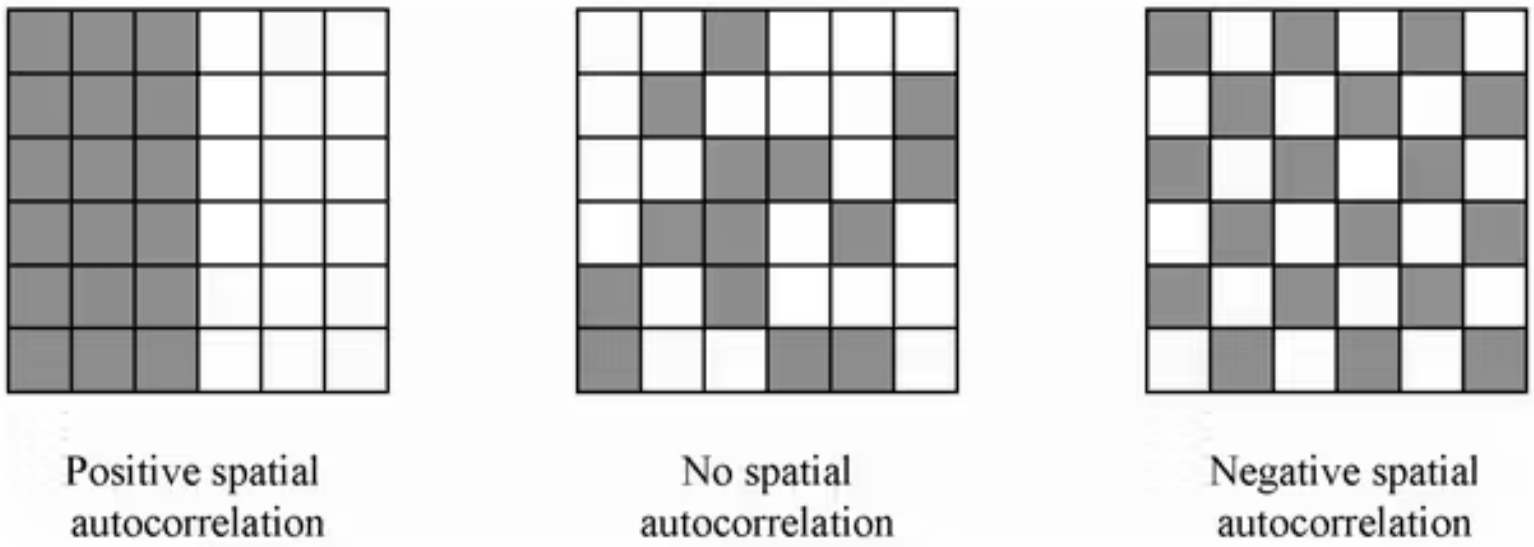
Babelon, I., 2015. Mapping place values for the green, compact and healthy city: Interlinking softGIS, sociotopes and communities of practice. *TRITA-LWR Degree Project*, *15*, p.118.

We have used spatial weights to measure how areas are similar to each other

Spatial autocorrelation measures how similar nearby locations are compared to locations farther apart.

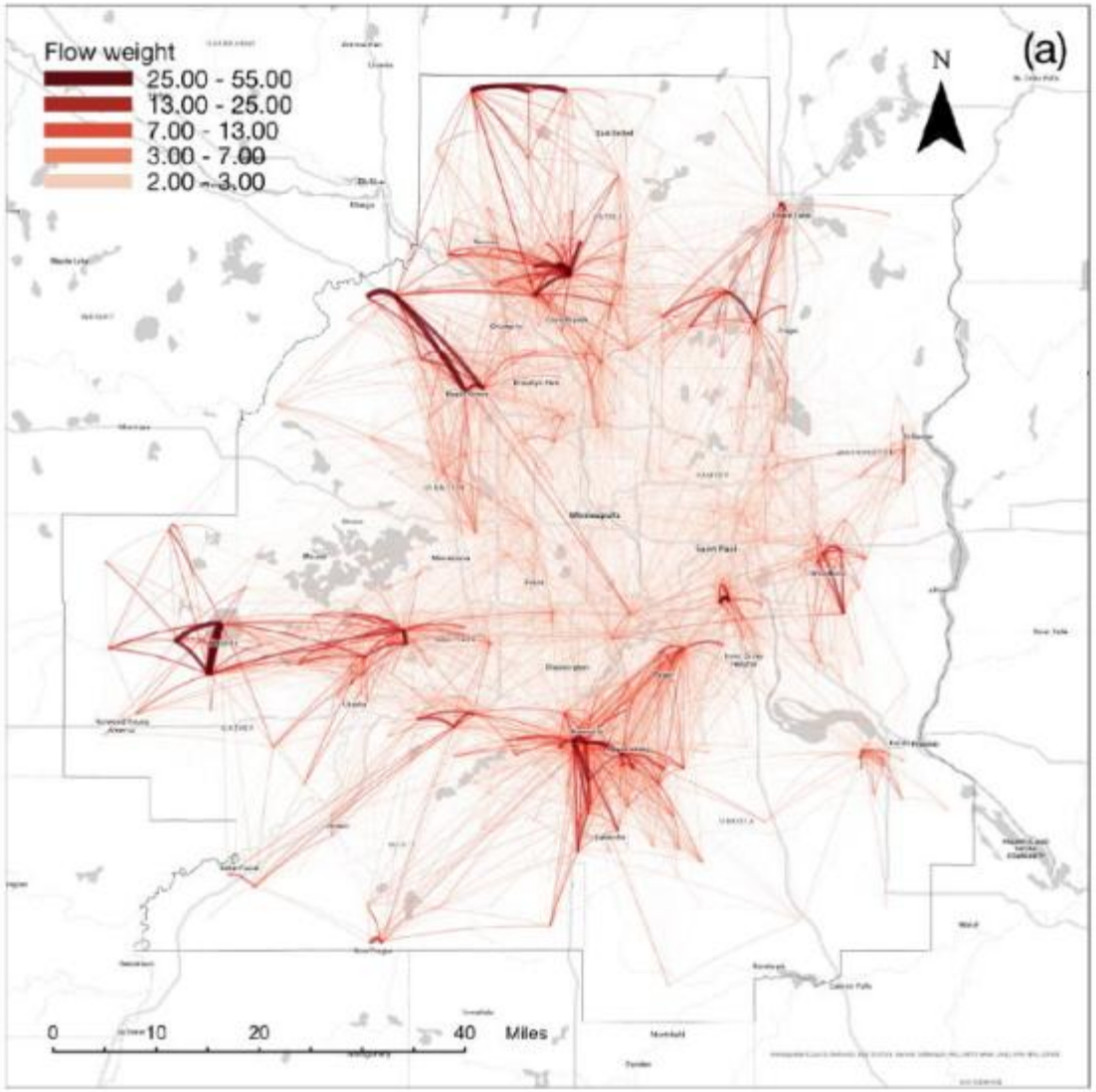
It answers one fundamental question:

Do values cluster in space, or are they randomly distributed?

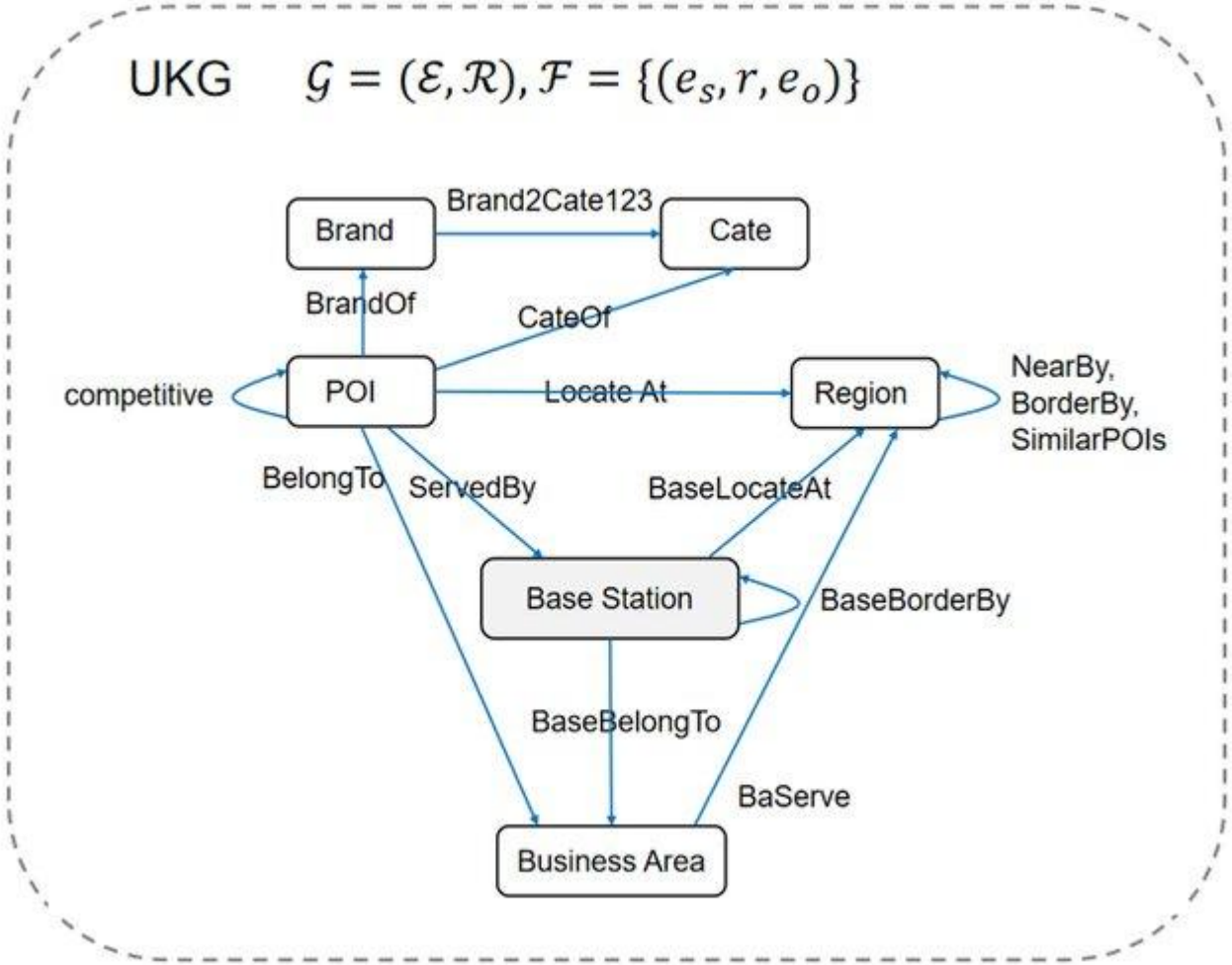


Source: André Egeli

Note! The high-level graph conceptualisation can be many things



Zhu, D. and Ma, Z., 2025. Gravity-informed deep flow inference for spatial evolution modeling in panel data. *International Journal of Geographical Information Science*, pp.1-29.

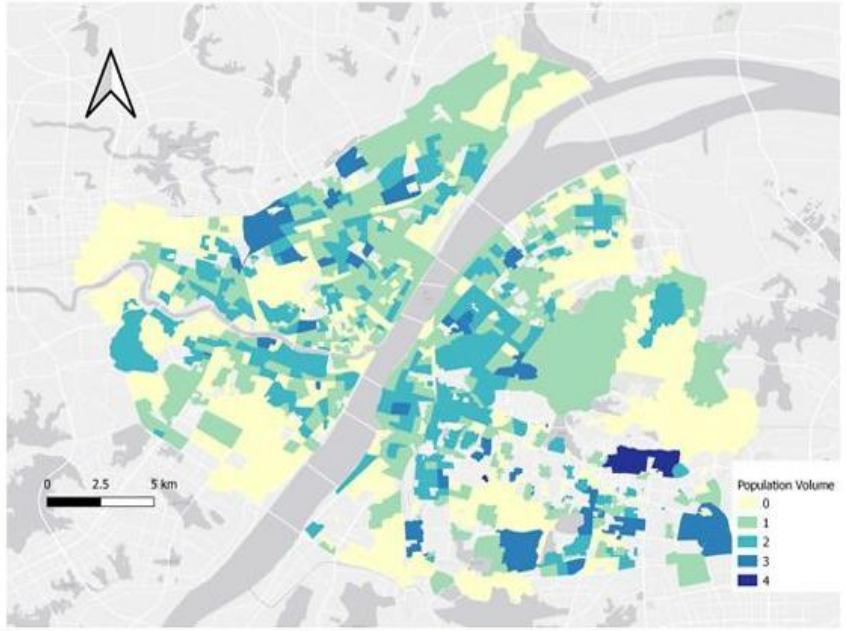


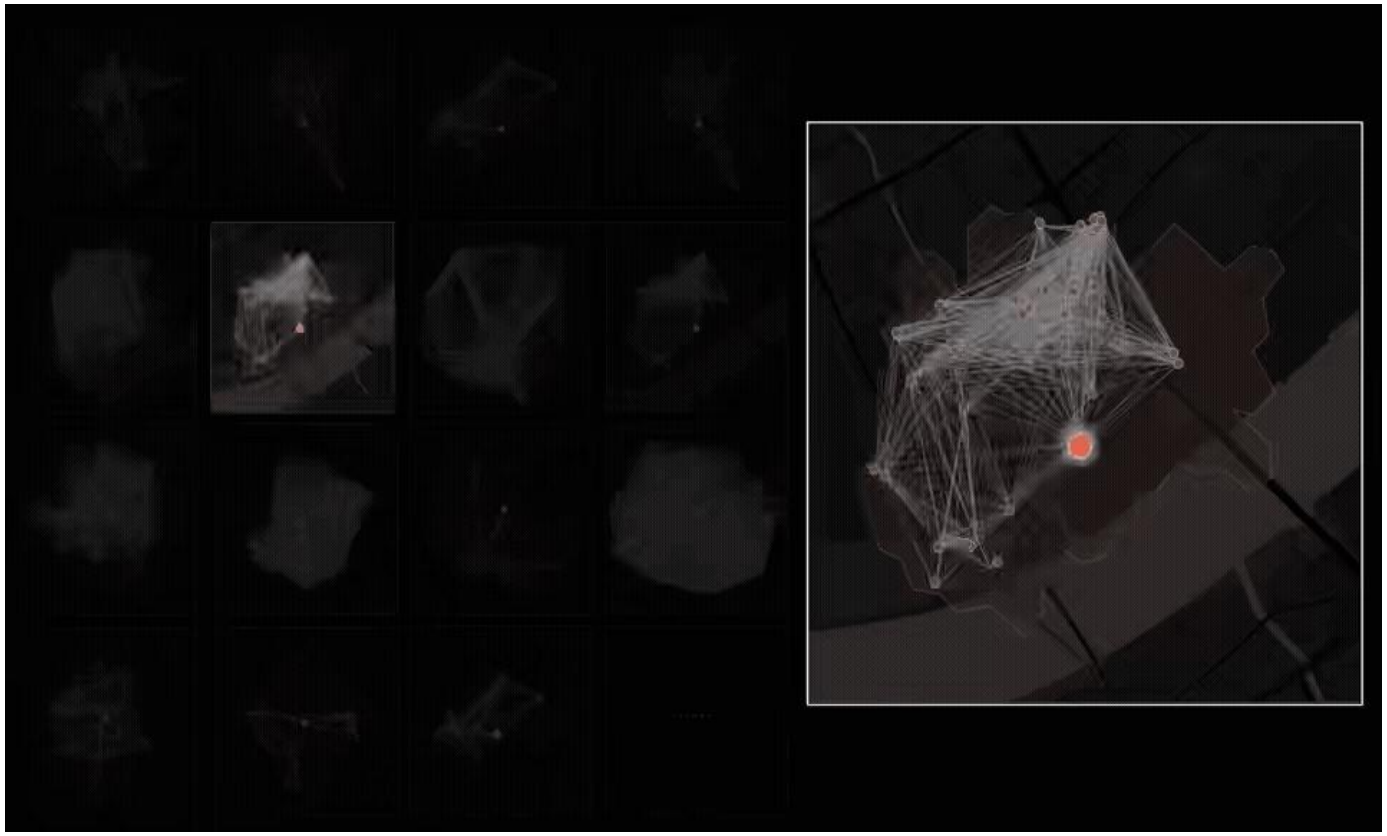
Liu, Z., Long, Q., Wang, H. and Li, Y., 2025, November. Multi-scale diffusion transformer for jointly simulating user mobility and mobile traffic pattern. In *Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems* (pp. 522-525).

Note! The high-level graph conceptualisation can be many things

Liu, P., Zhang, Y. and Biljecki, F., 2024. Explainable spatially explicit geospatial artificial intelligence in urban analytics. *Environment and Planning B: Urban Analytics and City Science*, *51*(5), pp.1104-1123.

Note! The high-level graph conceptualisation can be many things

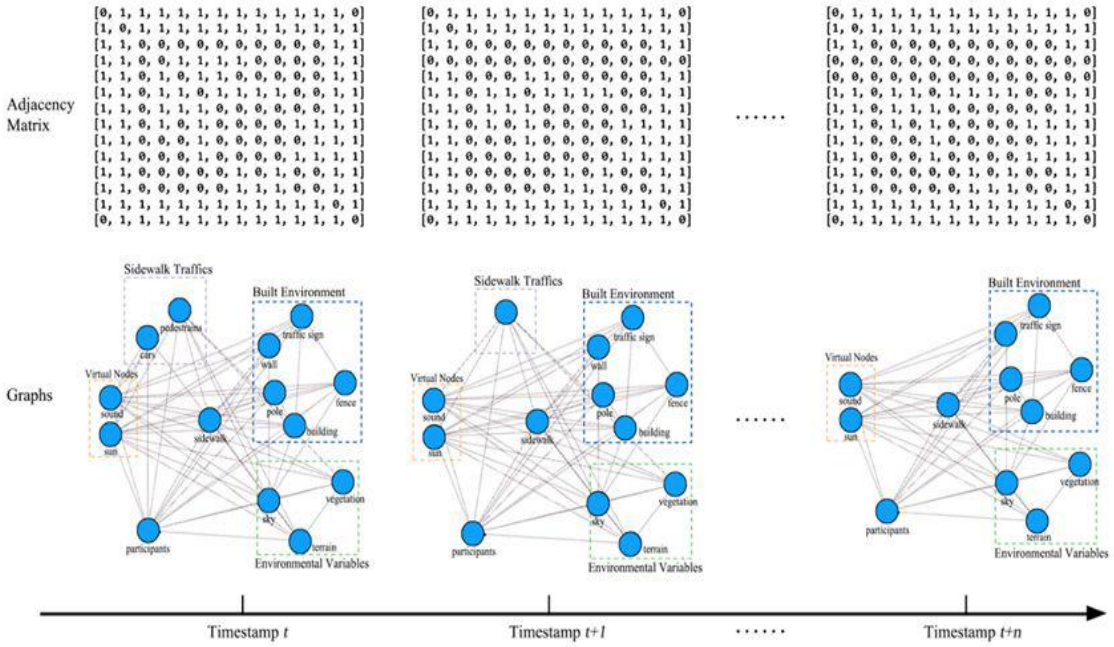


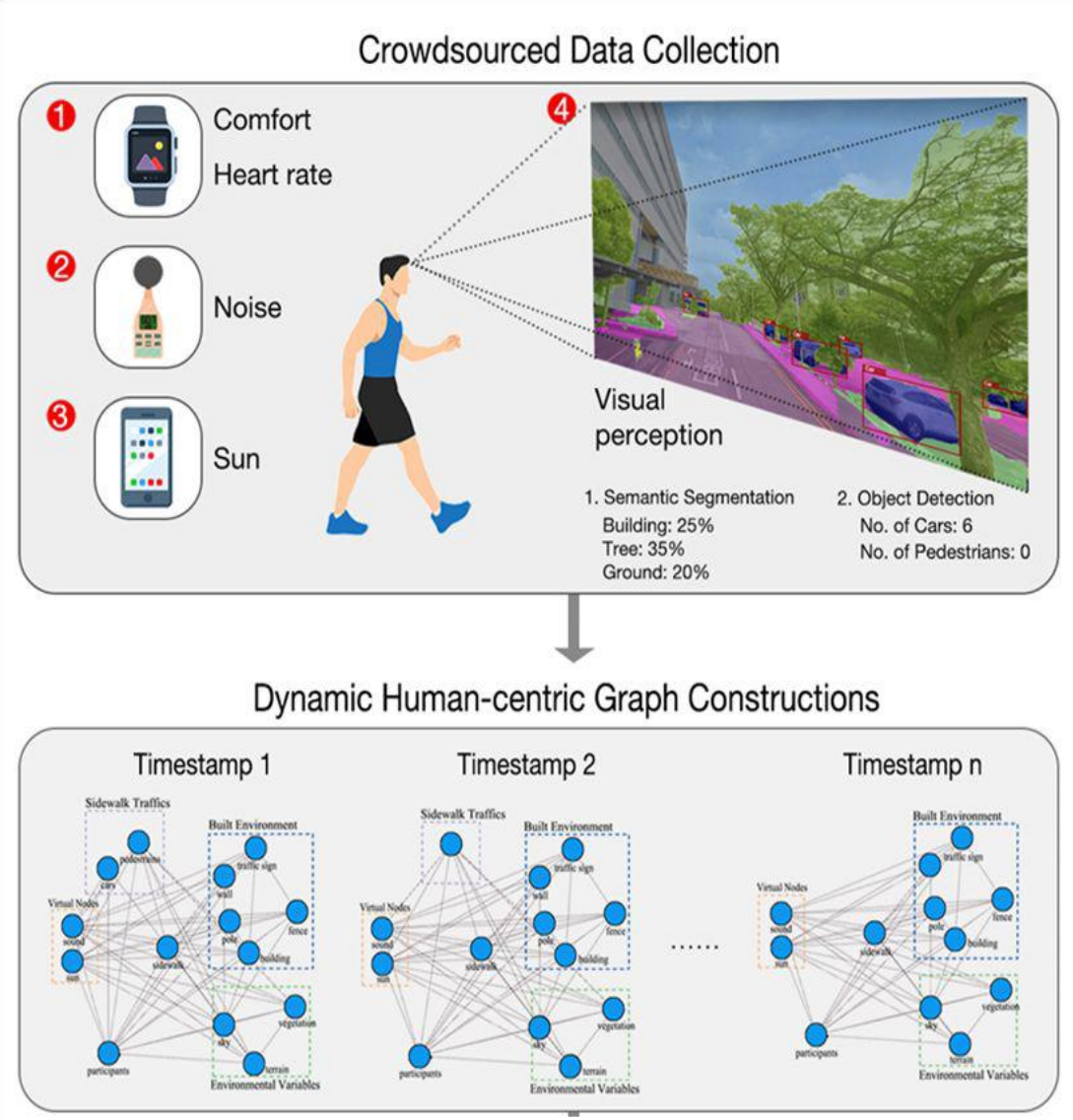


Liu, P., Zhang, Y. and Biljecki, F., 2024. Explainable spatially explicit geospatial artificial intelligence in urban analytics. *Environment and Planning B: Urban Analytics and City Science*, *51*(5), pp.1104-1123.

Note! The high-level graph conceptualisation can be many things

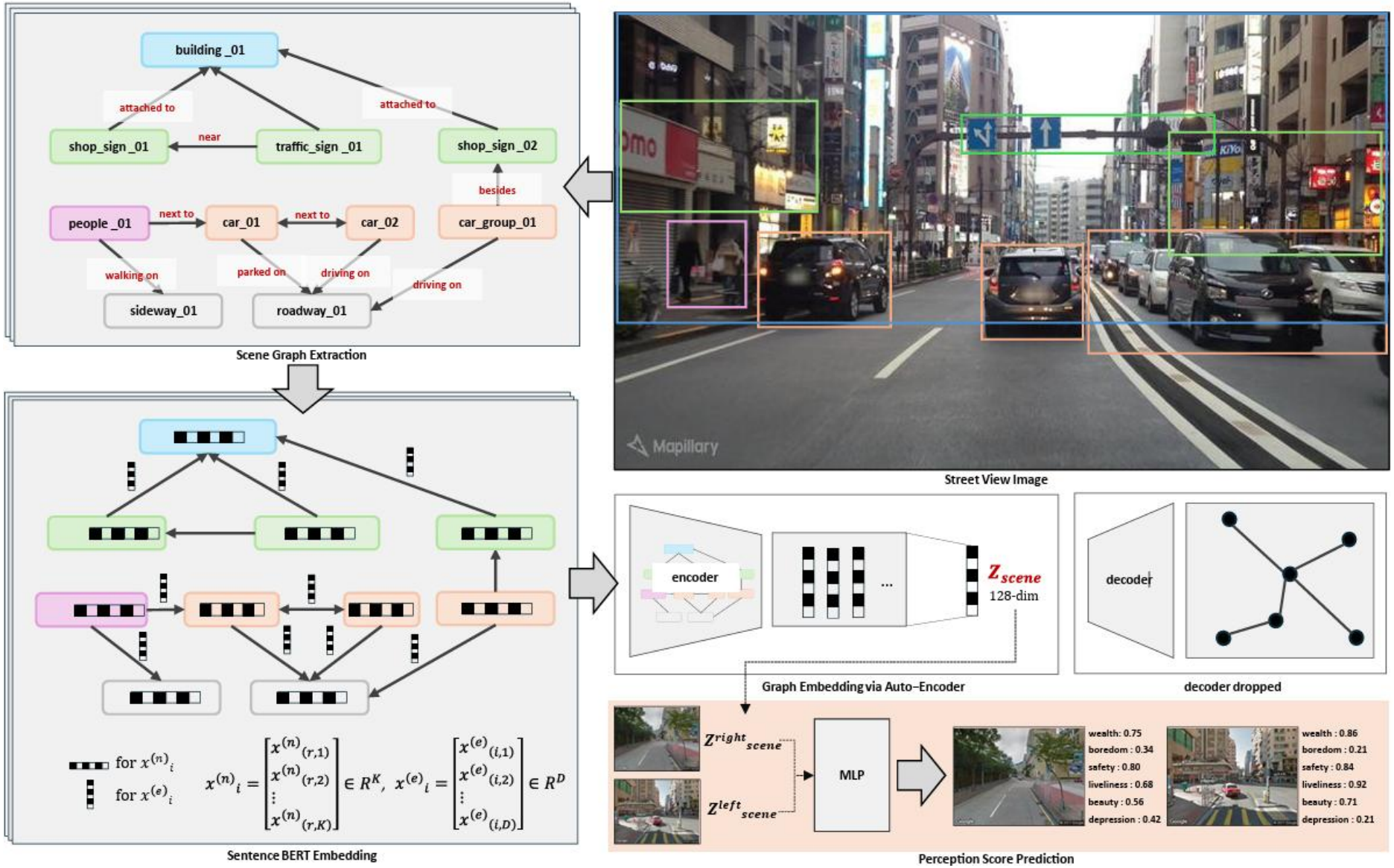
- Conceptualisation of the human movements and the interactions with surrounding urban features as dynamic graphs





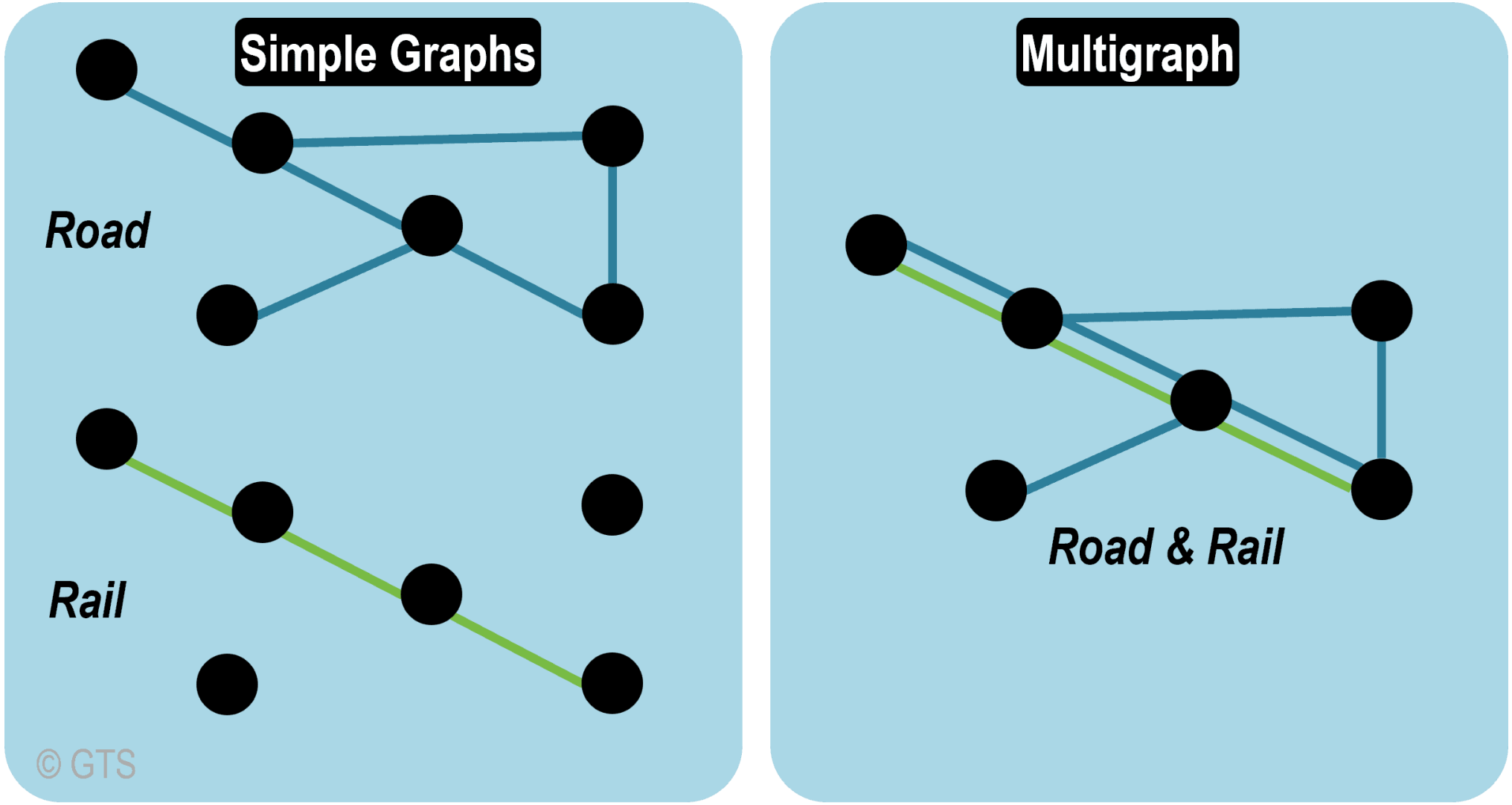
Liu, P., Zhao, T., Luo, J., Lei, B., Frei, M., Miller, C. and Biljecki, F., 2023. Towards human-centric digital twins: Leveraging computer vision and graph models to predict outdoor comfort. *Sustainable Cities and Society*, *93*, p.104480.

Note! The high-level graph conceptualisation can be many things



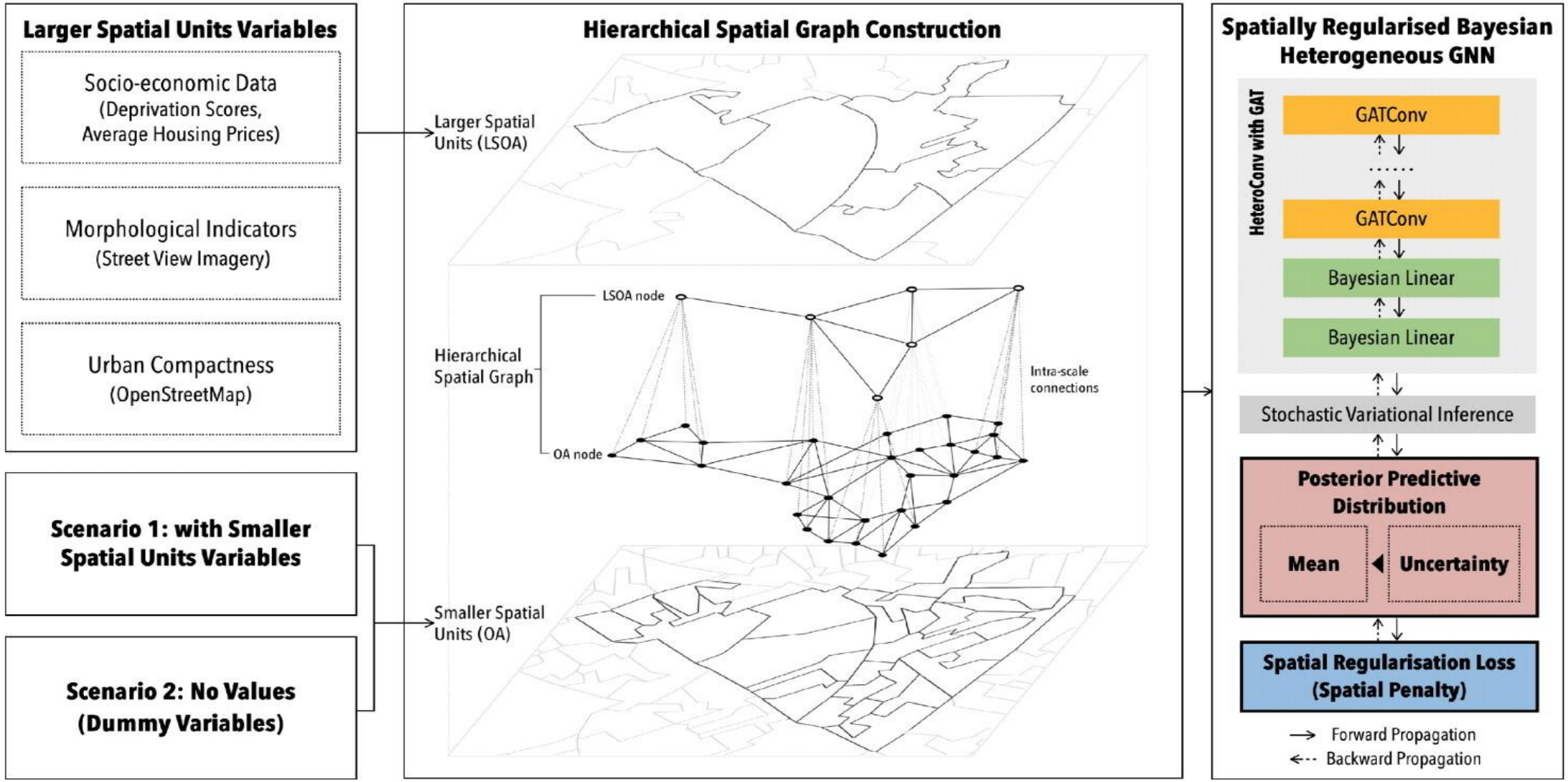
Liu, Y., Li, S., Liu, P., Zhang, Y. and Stouffs, R., 2025. From Pixels to Predicates Structuring urban perception with scene graphs. arXiv preprint arXiv:2512.19221.

Sometimes, we can stack graphs together



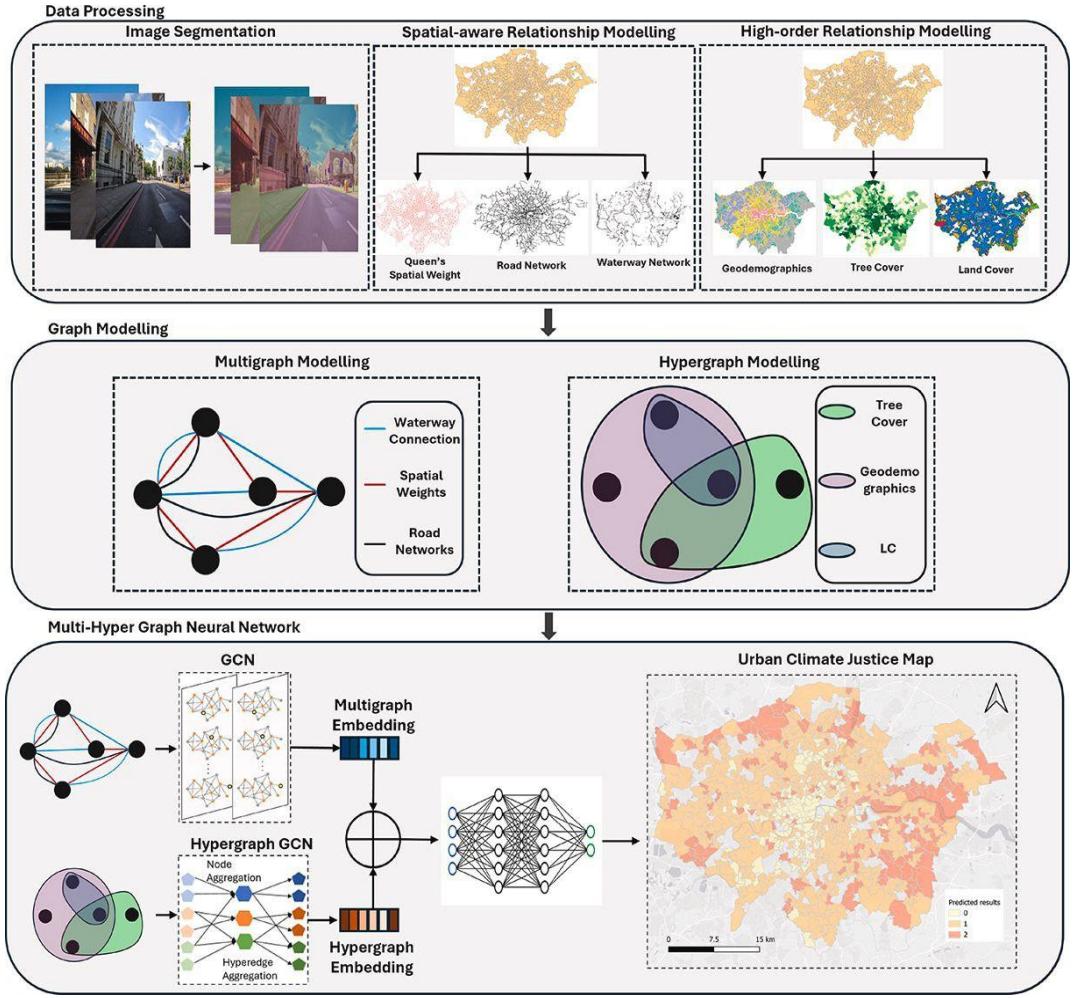
Source: https://transportgeography.org/contents/methods/graph-theory-definition-properties/simple-multi-graphs/

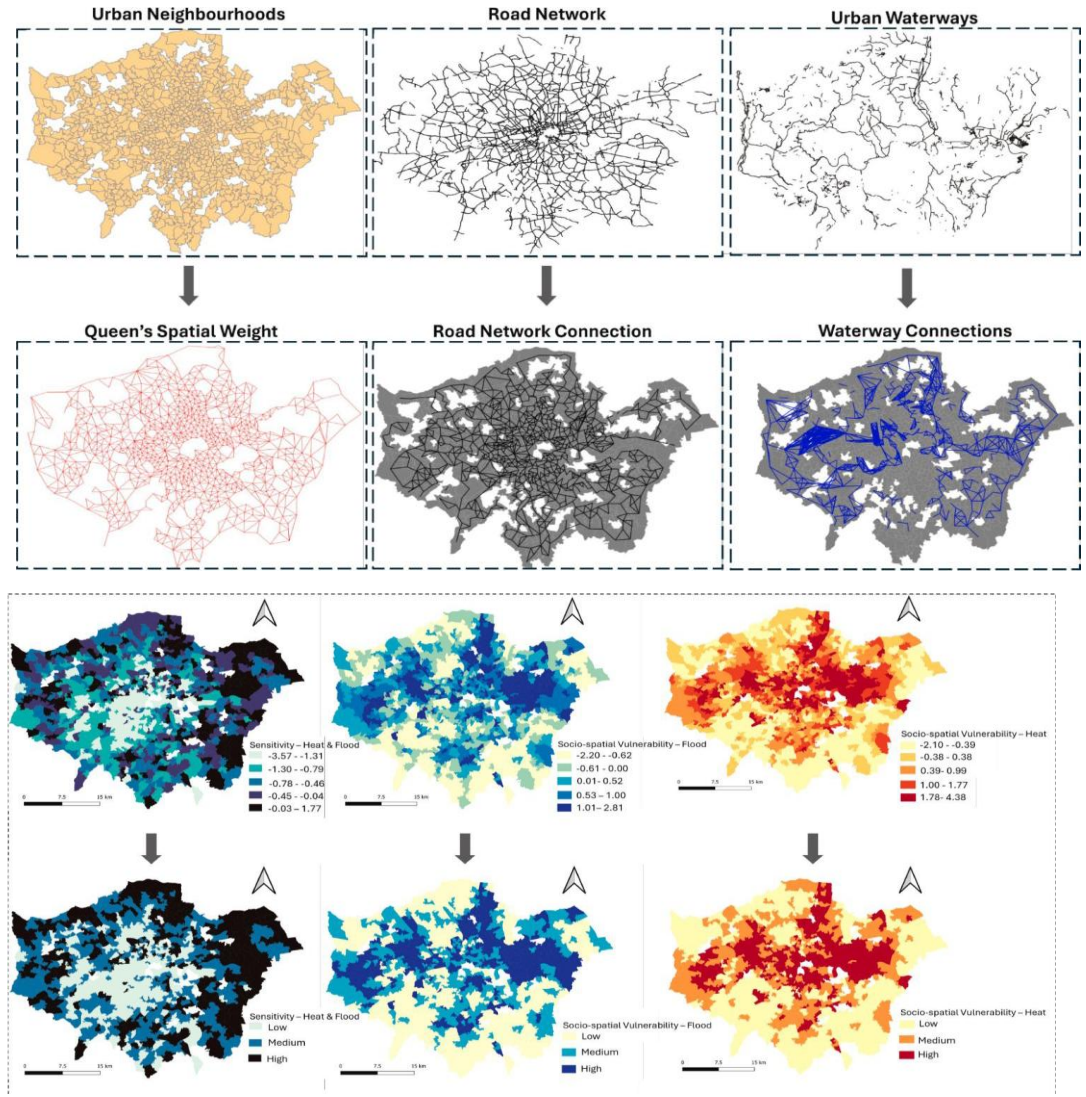
Sometimes, we can stack graphs together



Liu, P., Chen, Y., Liang, X., Li, H., Biljecki, F. and Stouffs, R., 2025. A graph neural network for small-area estimation: integrating spatial regularisation, heterogeneous spatial units, and Bayesian inference. *International Journal of Geographical Information Science*, pp.1-39.

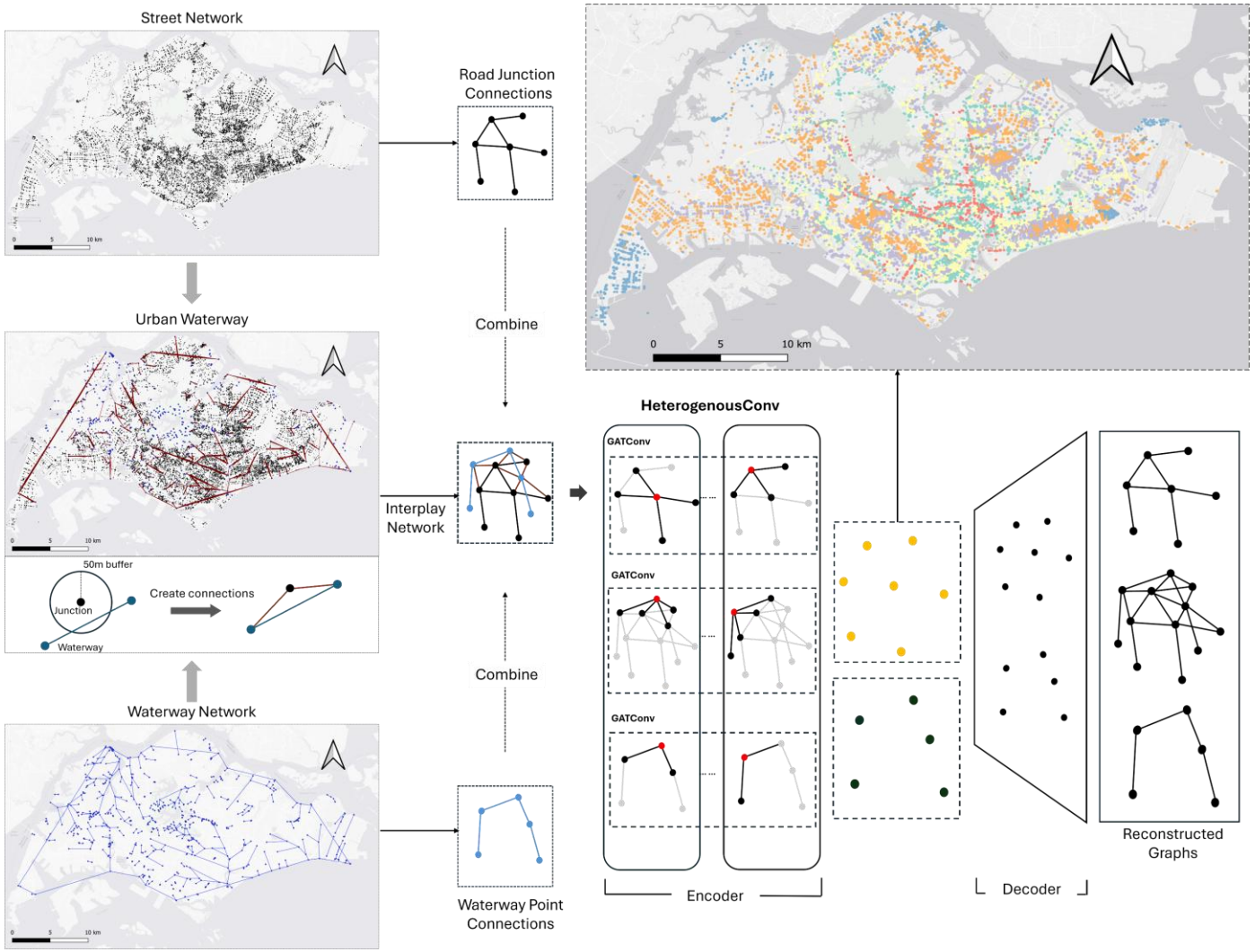
Sometimes, we can stack graphs together

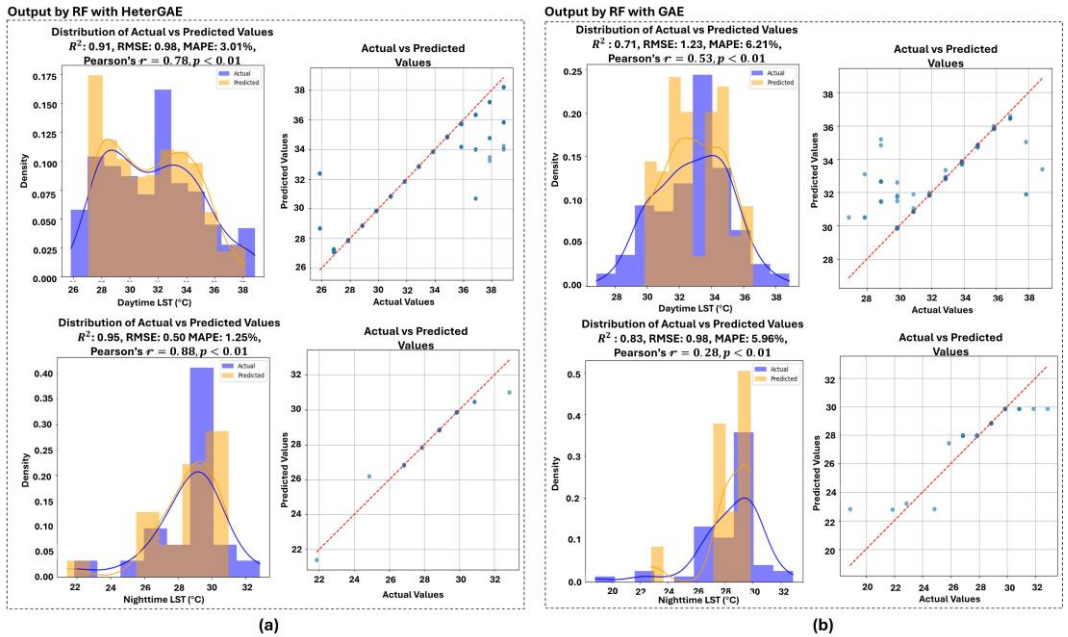


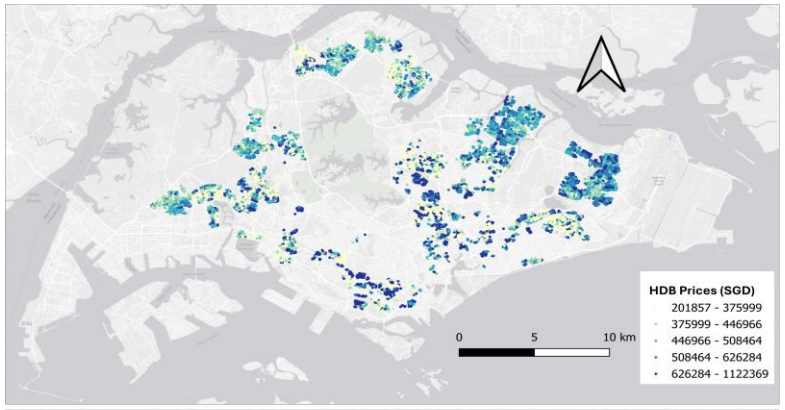


Liu, P., Lei, B., Huang, W., Biljecki, F., Wang, Y., Li, S. and Stouffs, R., 2025. Sensing climate justice: A multi-hyper graph approach for classifying urban heat and flood vulnerability through street view imagery. Sustainable Cities and Society, 118, p.106016.

Sometimes, we can stack graphs together







Metrics	HeterGAE with building-specific features	GAE with building-specific features	Only building-specific features	HeterGAE	GAE	GWR
RMSE	79400.60	85634.83	98007.19	138926.91	140214.45	79412.54
MAPE	8.88%	9.76%	10.79%	25.16%	26.23%	8.87%
R^2	0.74	0.69	0.65	0.33	0.35	0.73
Pearson's r	0.67***	0.52***	0.50***	0.18***	0.11***	0.66***

Liu, P., Wang, Y., Sabbata, S. D., Lei, B., Biljecki, F., Tang, J., & Stouffs, R. (2025). Living upon networks: A heterogeneous graph neural embedding integrating waterway and street systems for urban form understanding. Environment and Planning B: Urban Analytics and City Science.

Key Takeaways

- **City models** need to address and leverage a large number of different dimensions of knowledge, based on spatial knowledge
- At a high level of abstraction these structures can be described as **graphs with nodes** (land parcels, buildings) and edges (roads, supply lines).
- Spatial structures and geographic priors are key to be involved to develop *spatially-explicit models*